2:21 63%

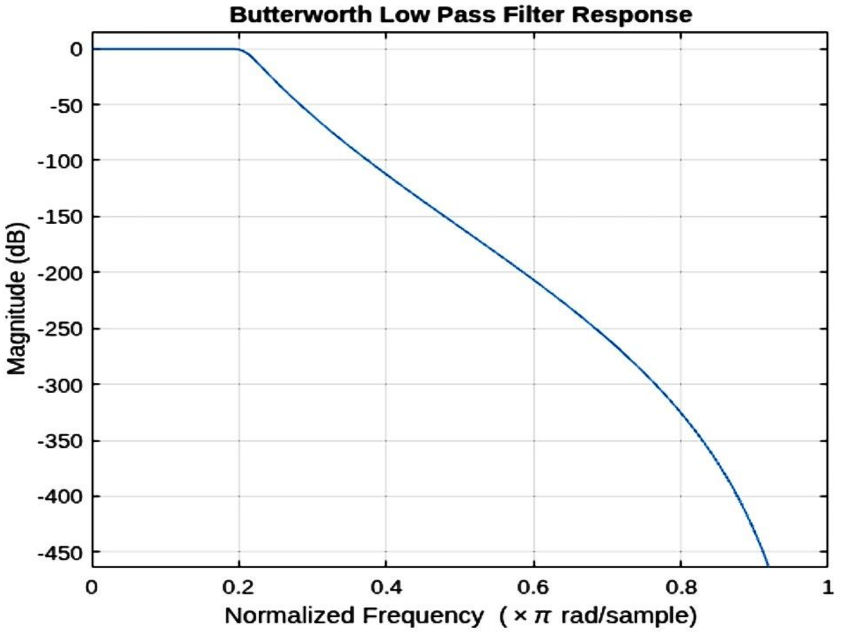
MATLAB Drive > Dsp1_1.m

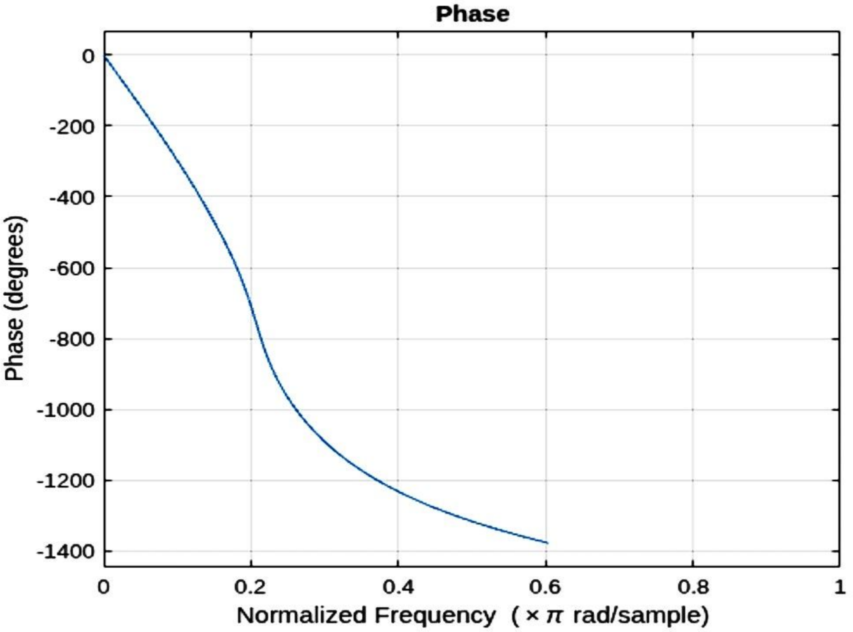
```matlab
clc;
clear;
close all;

% Specifications
Fs = 10000;          % Sampling frequency
Fp = 1000;           % Passband frequency
Fst = 1500;          % Stopband frequency
Rp = 1;              % Passband ripple
Rs = 60;             % Stopband attenuation

% Normalized Frequencies
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

% Filter Order Calculation
[N,Wn] = buttord(Wp,Ws,Rp,Rs);

% Butterworth LPF Design
[b,a] = butter(N,Wn,'low');

% Display Results
fprintf('Minimum Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.4f\n',Wn);

% Frequency Response
figure;
freqz(b,a);
title('Butterworth Low Pass Filter Response');

% Impulse Response
figure;
impz(b,a);
title('Impulse Response');

% Step Response
figure;
stepz(b,a);
title('Step Response');

% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');

% Pole-Zero Plot
figure;
```

2:20 63%

Figures SHOW DATA CURSOR

Figure 1: Dsp1_1





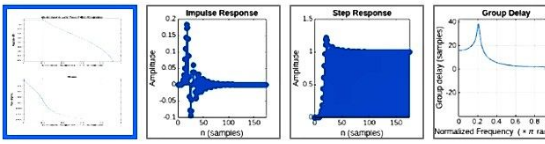


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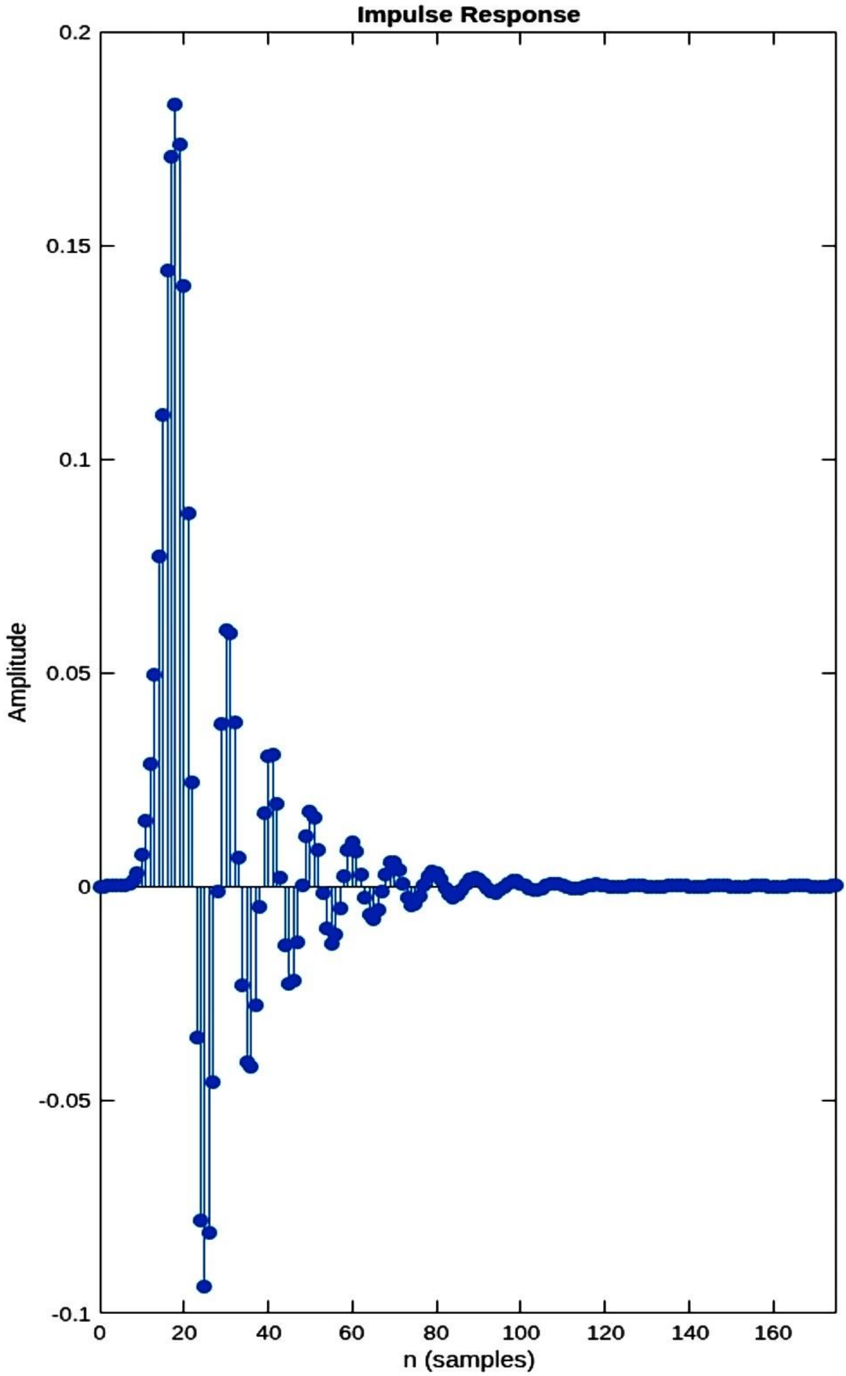
Figure 2: Dsp1_1



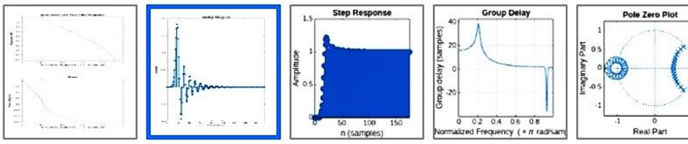


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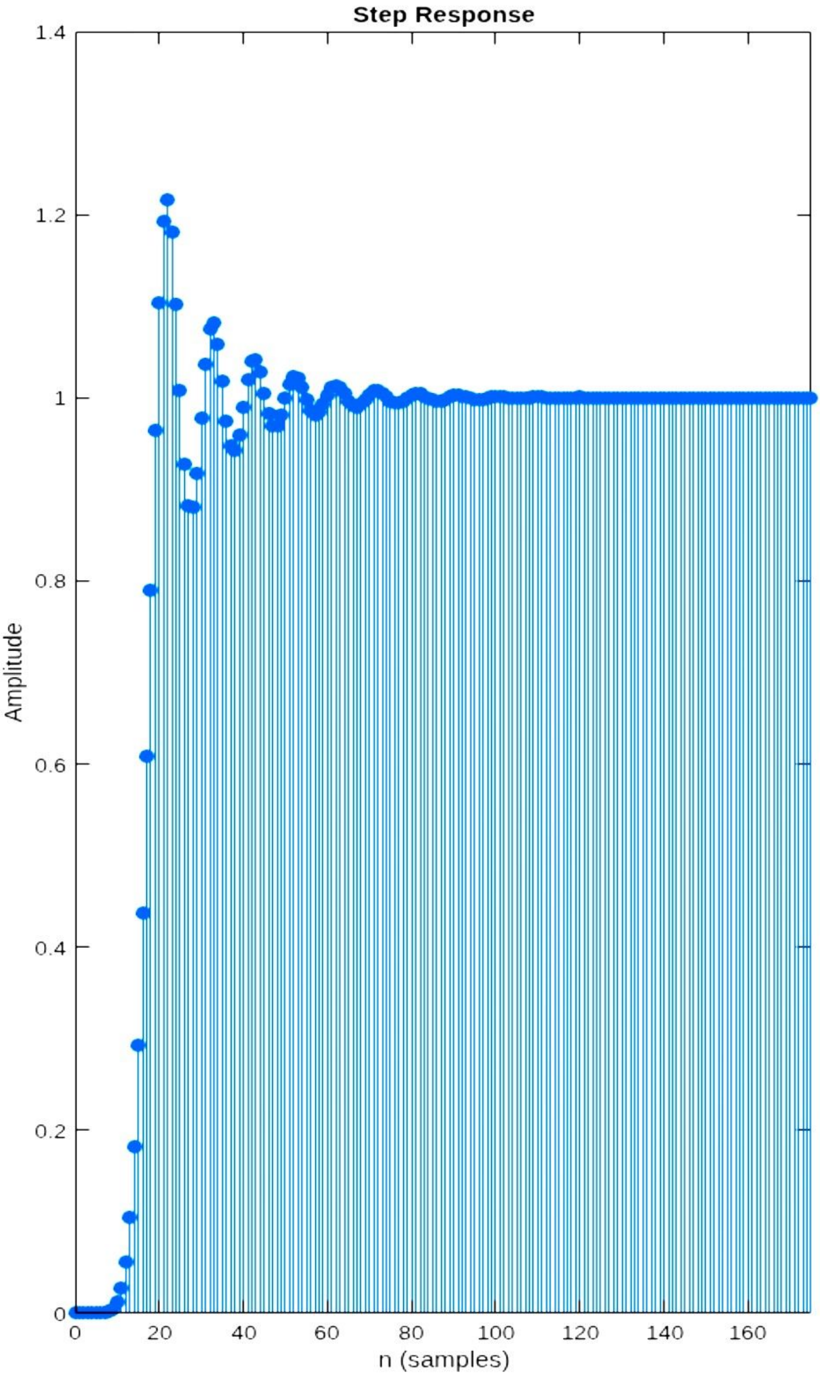
Figure 3: Dsp1_1



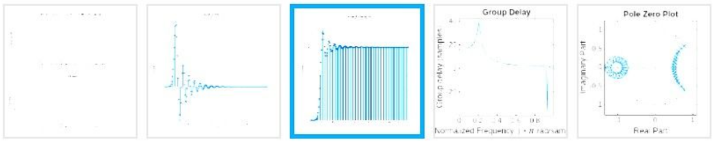


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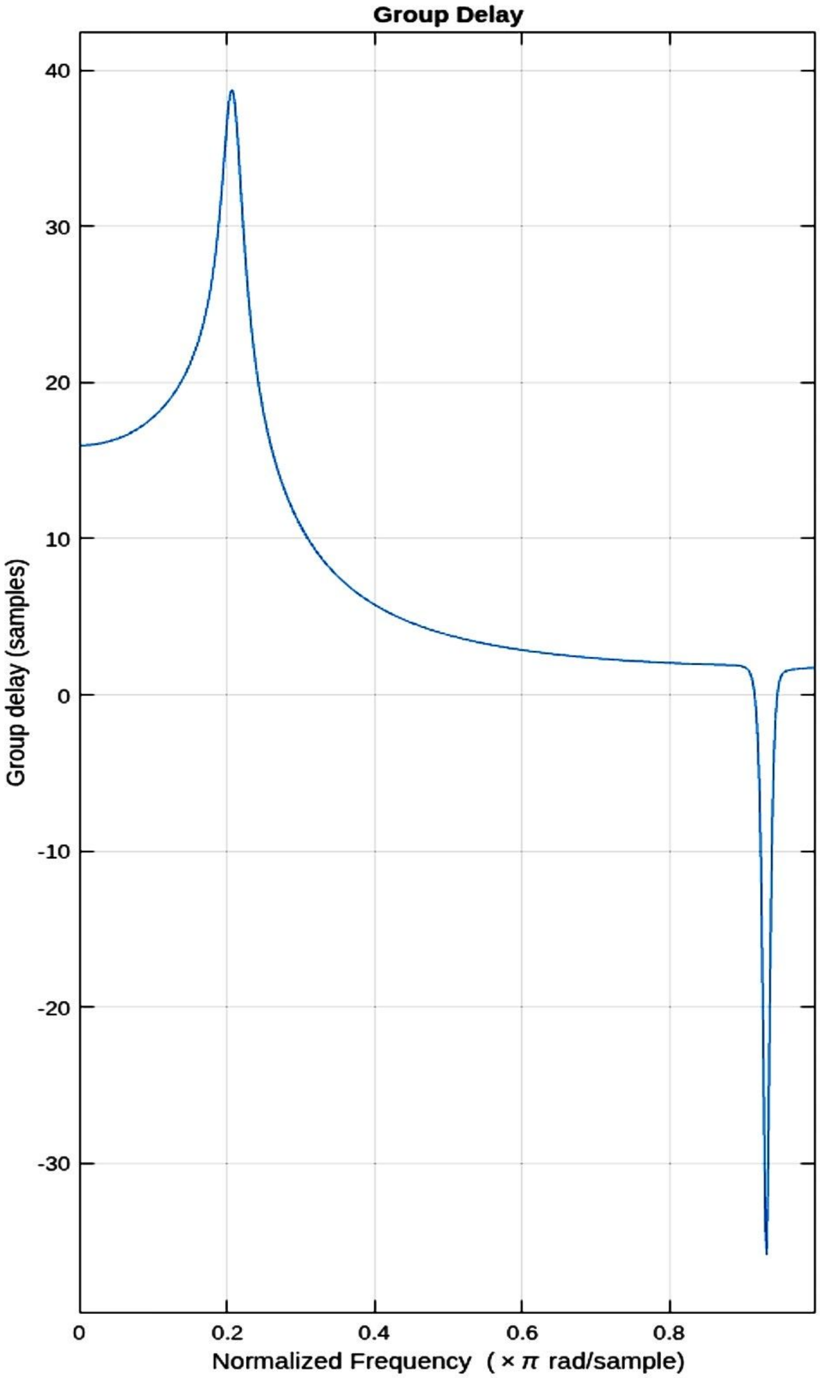
Figure 4: Dsp1_1



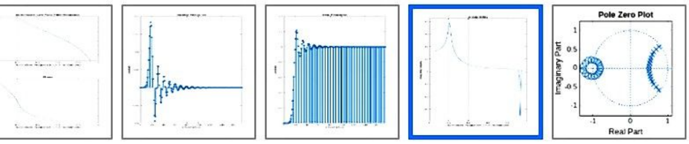
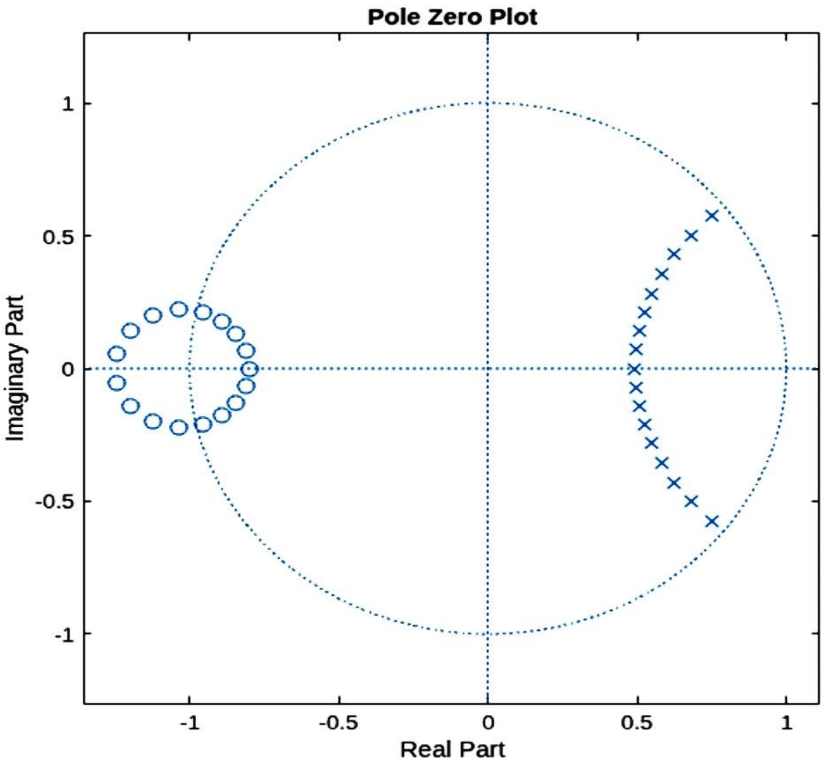
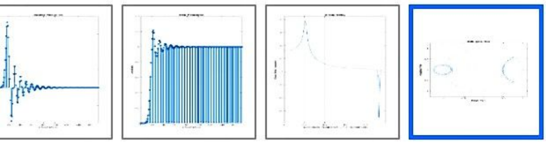


Figure 5: Dsp1_1





```
clc;
clear;
close all;

% Specifications
Fs = 10000;
Fp = 3000;
Fst = 2000;
Rp = 1;
Rs = 60;

% Normalization
Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

% Order Calculation
[N,Wn] = buttord(Wp,Ws,Rp,Rs);

% High Pass Filter Design
[b,a] = butter(N,Wn,'high');

fprintf('Minimum Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.4f\n',Wn);

% Frequency Response
figure;
freqz(b,a);
title('Butterworth High Pass Filter');

% Impulse Response
figure;
impz(b,a);
title('Impulse Response');

% Step Response
figure;
stepz(b,a);
title('Step Response');

% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');

% Pole Zero Plot
figure;
zplane(b,a);
```

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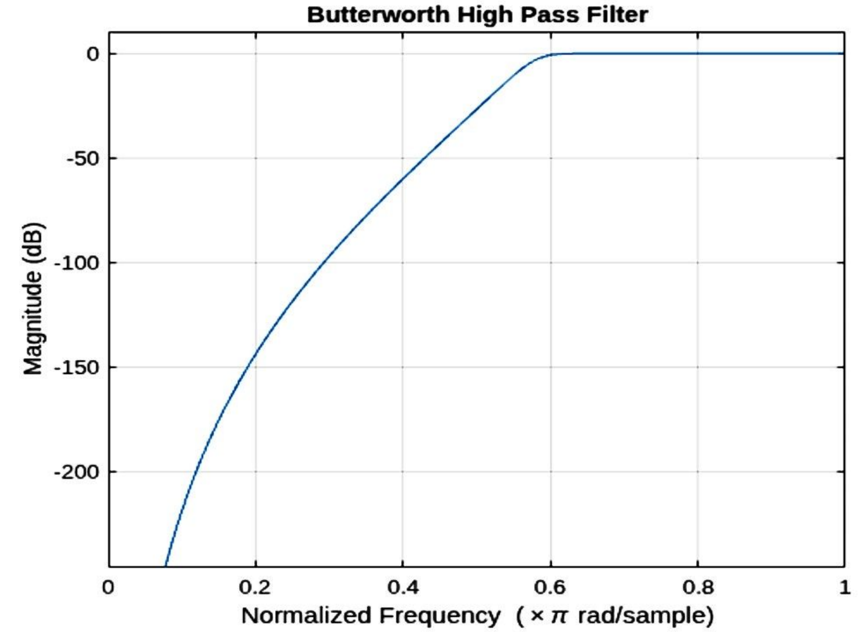
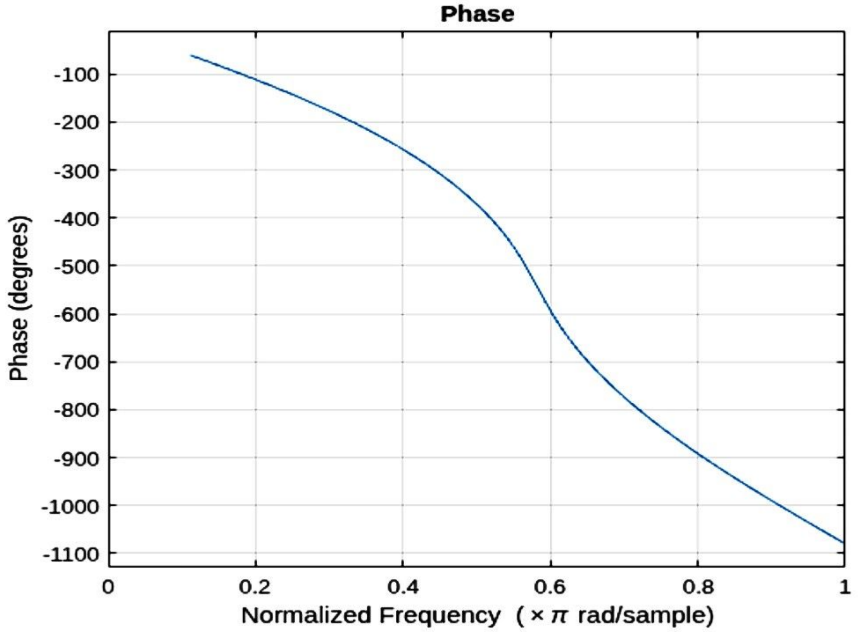
Figure 1: Dsp1_2





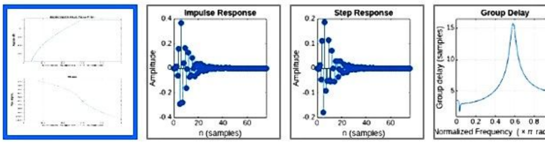


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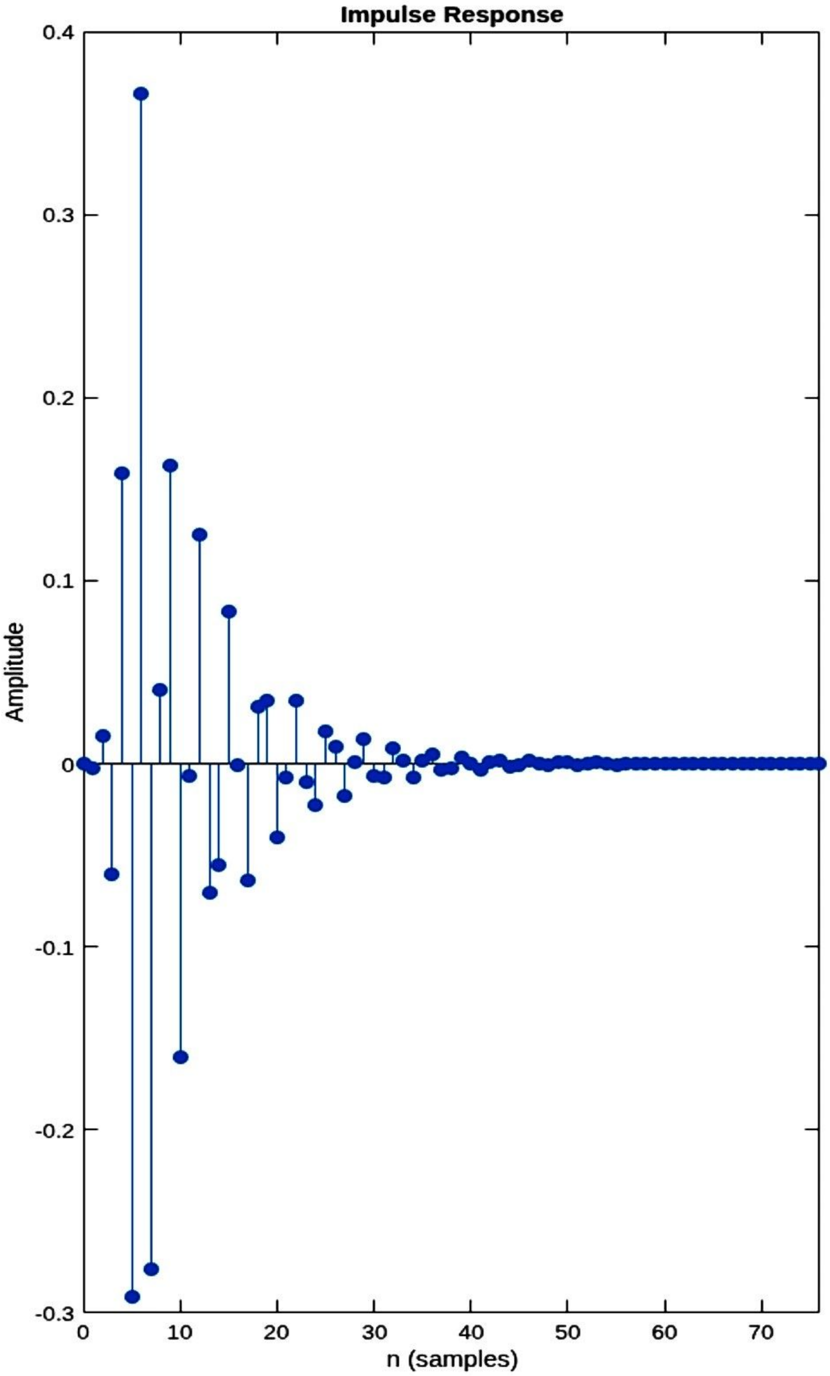
Figure 2: Dsp1_2



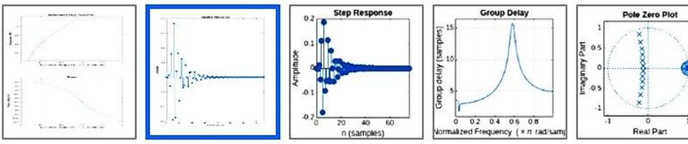


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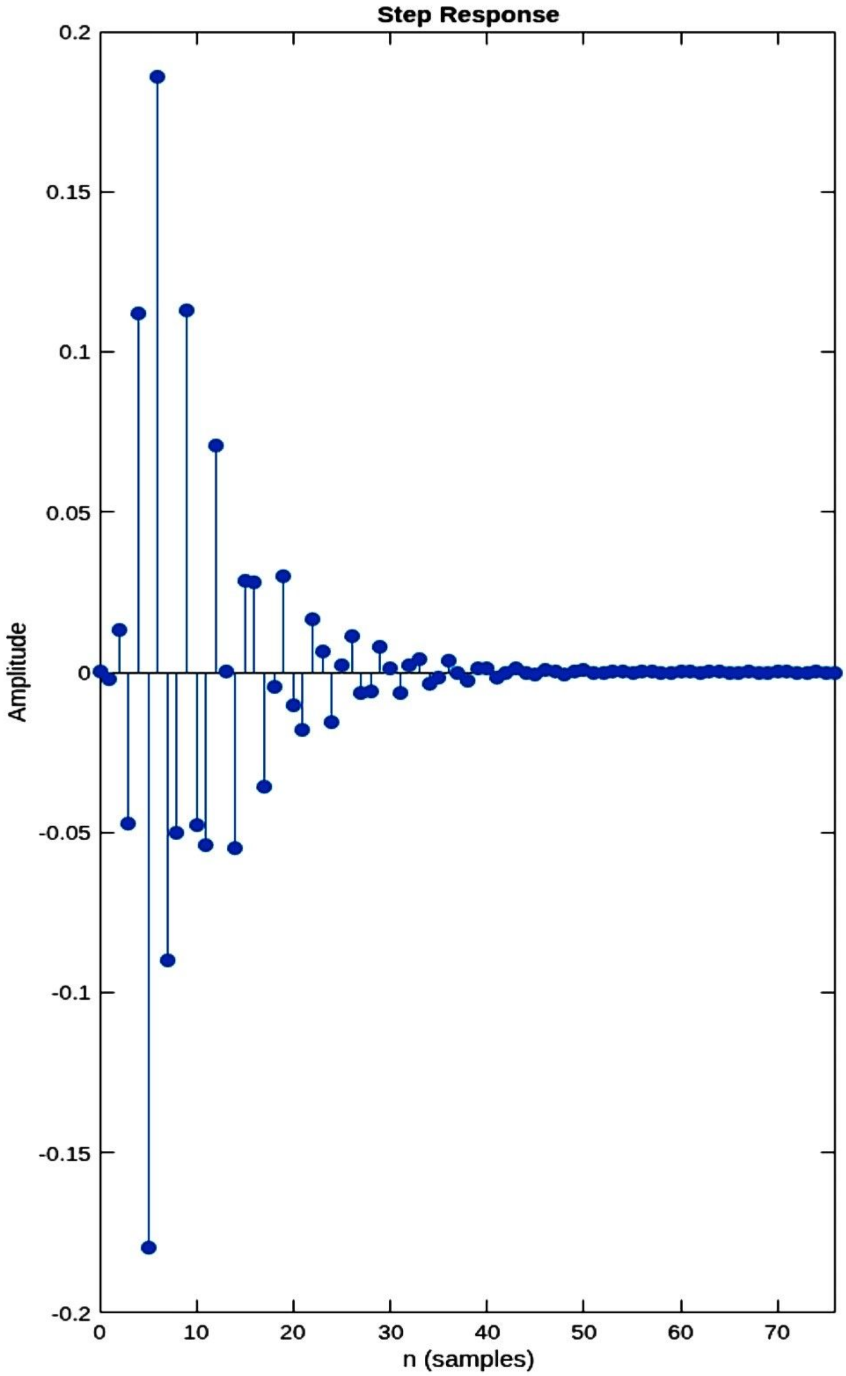
Figure 3: Dsp1_2



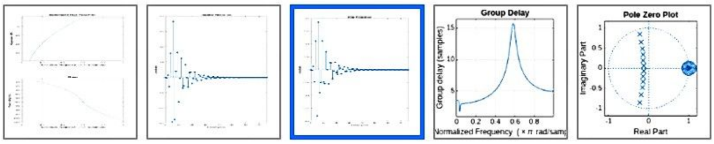


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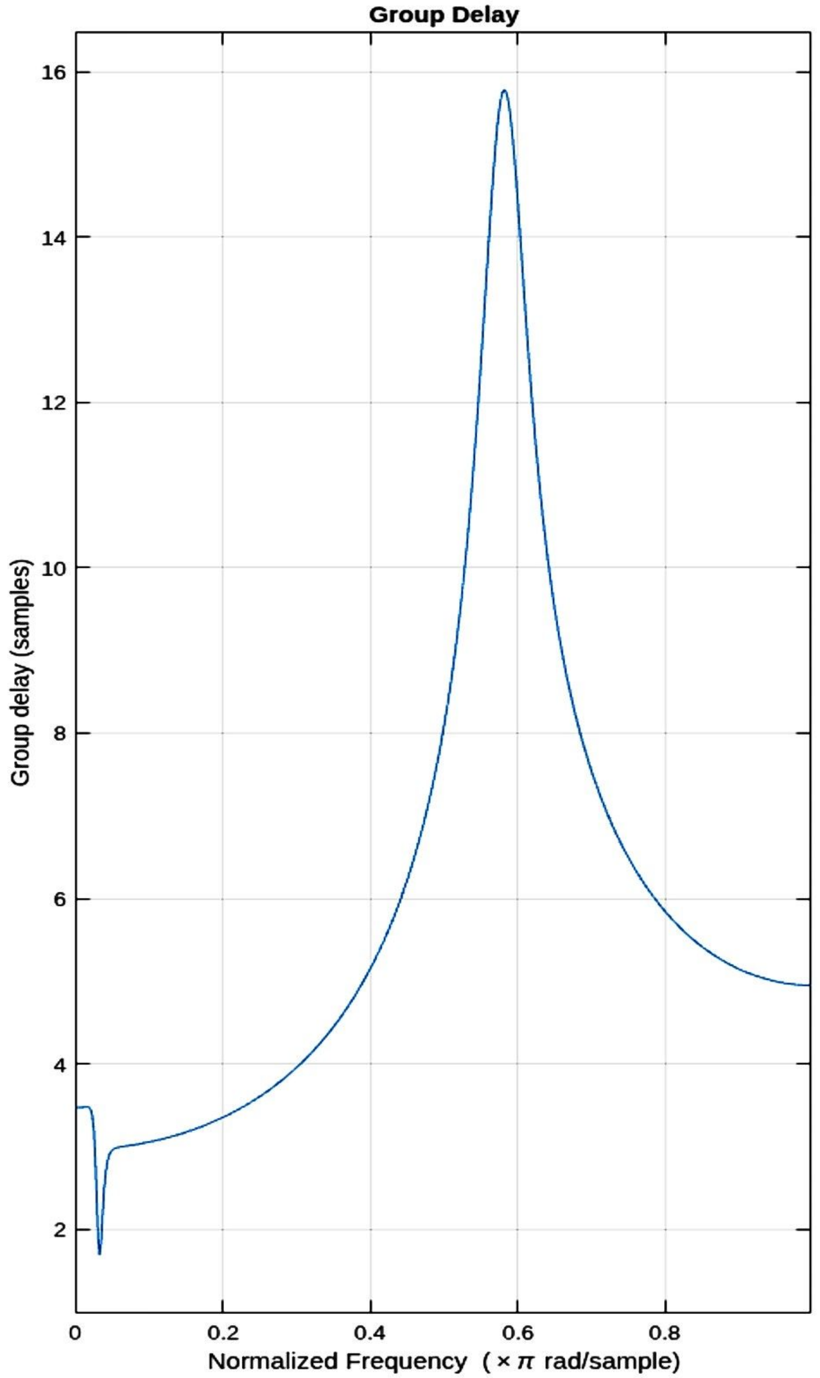
Figure 4: Dsp1_2



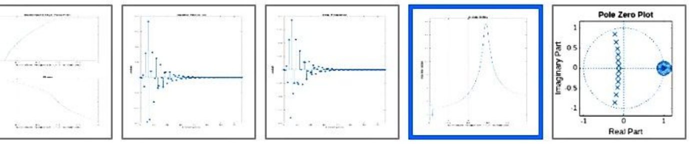


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Figure 5: Dsp1_2